The questions of this chapter

Linear algebra starts with two operations.

$$\text{Addition: } v + w, \qquad \text{Scalar multiplication: } cv.$$

Together they produce a linear combination

$$cv + dw.$$

A matrix records which coefficients multiply which column vectors.

$$Ax = x_1 a_1 + \cdots + x_n a_n.$$

The geometric question is which targets the combinations can reach. The algebraic question is which coefficients produce a specified target.

Manipulating vector coefficients and multiplying a matrix perform the same operation. Matrix multiplication is compact notation for a family of vector combinations.

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Vectors, components, and equality

A real vector with n components can be written as a column vector

$$v = (v_1, \dots, v_n)^T \in \mathbb{R}^n.$$

The superscript T transposes the horizontally written tuple into a column. Addition requires equal dimensions.

$$(v + w)_i = v_i + w_i.$$

Scalar multiplication acts component by component.

$$(cv)_i = cv_i.$$

The size n describes the ambient data dimension. It does not assert that a collection contains n independent directions. Span and rank answer that question.

T denotes transpose, not a variable. If the dimensions differ, $v + w$ is undefined.

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Zero vectors and zero coefficients

$$0 = \begin{bmatrix} 0 \\ \vdots \\ 0 \end{bmatrix}$$

is the identity for vector addition. The scalar 0 is the identity for scalar addition, whereas 1 is the identity for scalar multiplication. The two zero objects have different types despite sharing a symbol.

Vector subtraction also acts component by component.

$$v - w = v + (-1)w.$$

In the plane, an arrow represents a vector. For $p, q \in \mathbb{R}^n$, the displacement from p to q is $q - p$. An arrow from the origin to q represents q. Translating an arrow without changing its displacement leaves the vector unchanged.

The starting point of an arrow is not part of the free vector. The vector records its direction and magnitude.

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Linear combinations and span

For $v = (2, 1)^T$ and $w = (1, 2)^T$, the combinations are

$$cv + dw = \begin{bmatrix} 2c + d \\ c + 2d \end{bmatrix}.$$

To reach $b = (3, 3)^T$, solve

$$2c + d = 3, \qquad c + 2d = 3,$$

which gives $c = d = 1$.

The set of all $cv + dw$ is

$$\text{span}\{v, w\}.$$

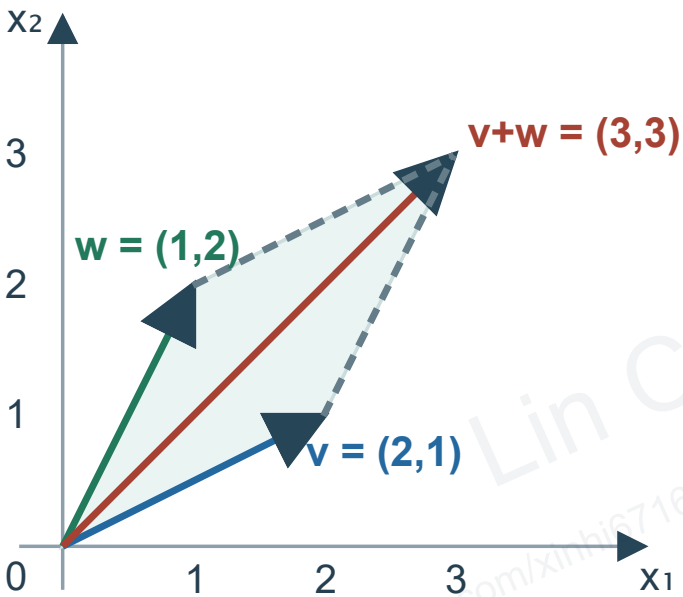
The coefficient matrix has determinant $2 \cdot 2 - 1 \cdot 1 = 3 \neq 0$. The directions are independent and their span is all of $\mathbb{R}^2$.

Geometrically, $v, w, v + w$ are two edges and the diagonal of the parallelogram with a vertex at the origin.

Span allows every real choice of coefficients. One fixed coefficient pair produces one vector, not a span.

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Linear combinations and span (continued)



v + w = (3,3)

(2,1) + (1,2)

= (2+1, 1+2)

= (3,3)

det[v w] = 3 ≠ 0

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Dependence and unreachable targets

If $w = 2v$, then $\text{span}\{v, w\} = \text{span}\{v\}$. For nonzero v in $\mathbb{R}^2$, this set is a line. If $v = 0$, the span is the zero subspace instead.

Take $v = (1, 2)^T$, $w = (2, 4)^T$, and $b = (3, 5)^T$. Every combination satisfies $y = 2x$, but $5 \neq 2 \cdot 3$. Consequently, the system

$$\begin{bmatrix} 1 & 2 \\ 2 & 4 \end{bmatrix} \begin{bmatrix} c \\ d \end{bmatrix} = \begin{bmatrix} 3 \\ 5 \end{bmatrix}$$

has no solution.

Suppose scalars $c_1, \ldots, c_k$ satisfy

$$c_1 v_1 + \cdots + c_k v_k = 0$$

with at least one $c_i \neq 0$. The vectors are linearly dependent. If no such nonzero coefficient vector exists, they are linearly independent.

Inconsistency is a geometric fact about a target outside the reachable subspace, rather than a failure of calculation.

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An equation test for independence

The vectors $v_1, \ldots, v_k$ are independent exactly when the homogeneous equation

$$Vc = 0, \qquad V = \begin{bmatrix} v_1 & \cdots & v_k \end{bmatrix}$$

has only the solution $c = 0$.

For a square matrix with $k = n$, columns that form a basis make V invertible, so $Vc = 0$ has only the zero solution. If $k > n$, elimination has at most n pivot variables and therefore at least one free variable. Setting that free variable nonzero produces a nonzero solution, so more than n vectors in $\mathbb{R}^n$ cannot be independent.

For example, three vectors in $\mathbb{R}^2$ necessarily give

$$c_1 v_1 + c_2 v_2 + c_3 v_3 = 0$$

a nonzero coefficient solution. Existence of such a homogeneous solution differs from existence of coefficients producing a particular b.

Separate the homogeneous equation $Vc = 0$ from consistency of $Vc = b$.

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Dot products and length

For $u, v \in \mathbb{R}^n$, the Euclidean inner product, or dot product, is

$$u \cdot v = u^T v = \sum_{i=1}^{n} u_i v_i.$$

The Euclidean norm is

$$\|v\|_2 = \sqrt{v^T v}.$$

A vector with $\|v\|_2 = 1$ is a unit vector. Normalize any nonzero v by forming

$$\frac{v}{\|v\|_2}.$$

For example, $v = (3, 4)^T$ has norm 5 and normalized vector $(3/5, 4/5)^T$.

In a complex vector space, the inner product $u^H v$ conjugates the left vector. Chapter 9 develops the complex formula.

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Angles and orthogonality

For nonzero real vectors u, v with angle θ,

$$\cos\theta = \frac{u^T v}{\|u\|_2 \|v\|_2}.$$

For nonzero vectors, $u^T v = 0$ is equivalent to $\theta = 90^\circ$. Such vectors are orthogonal. The zero vector is orthogonal to every vector because its inner product is zero, but it has no direction, so the angle formula does not apply.

For example,

$$u = (3, 4)^T, \quad v = (4, 3)^T, \quad \cos\theta = \frac{24}{5 \cdot 5} = \frac{24}{25}.$$

If $w = (4, -3)^T$, then $u^T w = 12 - 12 = 0$.

Orthogonal basis vectors must be nonzero. Orthogonality itself is defined by the inner product, including at zero.

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Cauchy–Schwarz and the triangle inequality

For every $u, v \in \mathbb{R}^n$,

$$|u^T v| \le \|u\|_2 \|v\|_2.$$

Proof. If $v = 0$, the result holds immediately. Otherwise let $t = (u^T v)/(v^T v)$. Then $0 \le \|u - tv\|^2 = u^T u - (u^T v)^2/(v^T v)$. Rearrangement proves the inequality. Equality holds exactly when $u = tv$. Squaring gives the equivalent statement

$$(u^T v)^2 \le (u^T u)(v^T v).$$

Equality holds exactly when u, v are linearly dependent. Applying $|u^T v| \le \|u\|\|v\|$ gives

$$\begin{aligned} \|u + v\|_2^2 &= \|u\|_2^2 + 2u^T v + \|v\|_2^2 \\ &\le (\|u\|_2 + \|v\|_2)^2. \end{aligned}$$

Thus $\|u + v\|_2 \le \|u\|_2 + \|v\|_2$. Equality requires the vectors to point in the same direction, or one to be zero. Arbitrary linear dependence does not suffice.

Taking square roots is valid because lengths are nonnegative. Preserve the direction condition when discussing equality.

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Matrix multiplication combines columns

Let

$$A = \begin{bmatrix} a_{11} & \cdots & a_{1n} \\ \vdots & & \vdots \\ a_{m1} & \cdots & a_{mn} \end{bmatrix} \in \mathbb{R}^{m \times n}.$$

For $x \in \mathbb{R}^n$,

$$Ax = x_1 \begin{bmatrix} a_{11} \\ \vdots \\ a_{m1} \end{bmatrix} + \cdots + x_n \begin{bmatrix} a_{1n} \\ \vdots \\ a_{mn} \end{bmatrix}.$$

The output belongs to $\mathbb{R}^m$. Two equivalent views compute Ax.

- The column view weights column i of A by x_i and adds the results.
- The row view takes the dot product of each row of A with x.

For example,

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Matrix multiplication combines columns (continued)

$$\begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix} \begin{bmatrix} 2 \\ -1 \end{bmatrix} = 2 \begin{bmatrix} 1 \\ 3 \end{bmatrix} - \begin{bmatrix} 2 \\ 4 \end{bmatrix} = \begin{bmatrix} 0 \\ 2 \end{bmatrix}.$$

Column combinations organize later chapters. The row view explains each equation separately.

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Two views of the same matrix–vector product

Let

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{bmatrix}, \qquad x = \begin{bmatrix} 1 \\ -1 \\ 2 \end{bmatrix}.$$

The matrix is 2×3: its input has three components and its output has two.

A : 2 × 3 **x : 3 × 1** **Ax : 2 × 1**

$$\begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{bmatrix} \times \begin{bmatrix} 1 \\ -1 \\ 2 \end{bmatrix} = \begin{bmatrix} 5 \\ 11 \end{bmatrix}$$

column $1 \cdot (1,4) - (2,5) + 2 \cdot (3,6) = (5,11)$

row

$1 \cdot 1 + 2 \cdot (-1) + 3 \cdot 2 = 5$

$4 \cdot 1 + 5 \cdot (-1) + 6 \cdot 2 = 11$

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Work through the columns and the rows

Column view: weight the three columns of A by the components of x:

$$Ax = 1\begin{bmatrix}1\\4\end{bmatrix} - 1\begin{bmatrix}2\\5\end{bmatrix} + 2\begin{bmatrix}3\\6\end{bmatrix} = \begin{bmatrix}1-2+6\\4-5+12\end{bmatrix} = \begin{bmatrix}5\\11\end{bmatrix}.$$

Row view: the first row dotted with x gives $1-2+6=5$; the second gives $4-5+12=11$. Both calculations follow from $(Ax)_i = \sum_{j=1}^{3} a_{ij}x_j$; they group the same scalar products in different ways.

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Three readings of $Ax = b$

$$Ax = b$$

The equation asks any of the following equivalent questions.

1. Which linear combination of the columns produces b?
2. Which x has dot product b_i with every row i?
3. Does the linear map $x \mapsto Ax$ send an input to b?

For $b = (0, 2)^T$ and the preceding matrix, solve

$$x_1 + 2x_2 = 0, \qquad 3x_1 + 4x_2 = 2.$$

The first equation gives $x_1 = -2x_2$. Substitution yields $-2x_2 = 2$, hence $x = (2, -1)^T$.

The shared answer $x = (2, -1)^T$ follows because the three readings describe the same equality.

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Unique and infinitely many solutions

A square matrix $A \in \mathbb{R}^{n \times n}$ has a unique solution for every b exactly when $Ax = 0$ has only $x = 0$. In that case A is invertible. Indeed, no free variables means n pivots; elimination then solves every right-hand side uniquely.

If a nonzero z satisfies $Az = 0$ and x_p solves $Ax = b$, then

$$A(x_p + cz) = b + cAz = b$$

for every scalar c. Therefore the solution is not unique.

For example,

$$\begin{bmatrix} 1 & 2 \\ 2 & 4 \end{bmatrix} x = 0$$

has solutions $x_2 = t,\ x_1 = -2t$, which form a line.

First identify the homogeneous solutions. Chapter 3 expresses the complete solution as a particular solution plus the nullspace.

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A 2×2 inverse and its conditions

For

$$A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}, \quad ad - bc \neq 0,$$

the inverse is

$$A^{-1} = \frac{1}{ad - bc} \begin{bmatrix} d & -b \\ -c & a \end{bmatrix}.$$

For example,

$$A = \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}, \quad A^{-1} = \frac{1}{3} \begin{bmatrix} 2 & -1 \\ -1 & 2 \end{bmatrix}.$$

Direct multiplication verifies

$$\frac{1}{3} \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix} \begin{bmatrix} 2 & -1 \\ -1 & 2 \end{bmatrix} = I_2.$$

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A 2×2 inverse and its conditions (continued)

Check the nonzero determinant condition before using this inverse. For numerical solutions, direct elimination usually costs less than constructing A^{-1}. A^{-1} is a matrix. The number $1/\det A$ is a scalar.

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Review and common misconceptions

Check that you can explain the following distinctions.

- A combination $cv + dw$ and the set $\text{span}\{v, w\}$.
- The column and row views of $Ax = b$.
- Nonzero homogeneous solutions, infinitely many solutions, and inconsistency.
- Zero dot product and linear dependence.
- Solvability for one b and solvability for every b.

Final check. If $\text{span}\{v, w\} = \mathbb{R}^2$, must v, w be independent? Yes. A nontrivial dependence would make one vector a multiple of the other, or one vector zero, leaving a span of dimension at most one. That contradicts spanning the plane.

Chapter 2 turns these decisions into systematic elimination.

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Coordinates in polynomial and matrix spaces

Prerequisites: vector spaces, linear independence, and bases.

Vector operations also apply to spaces of functions and matrices.

In $\mathcal{P}_2$, use the monomial basis $B = (1, t, t^2)$ and define

$$p_1 = 1 + t, \qquad p_2 = 1 - t, \qquad p_3 = 1 + t^2.$$

The coordinate map is a linear isomorphism, so

$$[p_1]_B = (1, 1, 0)^T, \quad [p_2]_B = (1, -1, 0)^T, \quad [p_3]_B = (1, 0, 1)^T.$$

Using these coordinates as columns gives

$$D = \begin{bmatrix} 1 & 1 & 1 \\ 1 & -1 & 0 \\ 0 & 0 & 1 \end{bmatrix}, \qquad \det D = (1 \cdot (-1) - 1 \cdot 1) \cdot 1 = -2 \neq 0.$$

Thus p_1, p_2, p_3 form a basis of $\mathcal{P}_2$. For $q(t) = 3 + 2t + t^2$, solve $Dc = [q]_B = (3, 2, 1)^T$, obtaining $c = (2, 0, 1)^T$. Hence

$$q = 2p_1 + p_3.$$

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Coordinates in polynomial and matrix spaces (continued)

The same construction applies to matrices. Use column-major coordinates on $M_2(\mathbb{R})$.

$$\begin{bmatrix} a & b \\ c & d \end{bmatrix} \longmapsto (a, c, b, d)^T.$$

Let $A_1 = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}$、$A_2 = \begin{bmatrix} 1 & 1 \\ 1 & 0 \end{bmatrix}$、$A_3 = \begin{bmatrix} 0 & 1 \\ 1 & 1 \end{bmatrix}$. Their coordinate columns are

$$\begin{bmatrix} 1 & 1 & 0 \\ 0 & 1 & 1 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{bmatrix}.$$

If $c_1 A_1 + c_2 A_2 + c_3 A_3 = 0$, comparison of entries gives $c_1 + c_2 = 0$, $c_2 + c_3 = 0$, and $c_3 = 0$. All three coefficients vanish, proving independence. An invertible coordinate map lets us test a basis even when its elements are functions or matrices.

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Polynomial spectral mapping without diagonalization

Prerequisites: eigenvalues, determinants, and polynomial factorization over the complex numbers.

Let $r(t) = \sum_{k=0}^{d} c_k t^k$, with $Av = \lambda v$ and $v \neq 0$. Since $A^k v = \lambda^k v$,

$$r(A)v = \sum_{k=0}^{d} c_k A^k v = \sum_{k=0}^{d} c_k \lambda^k v = r(\lambda)v.$$

Every $r(\lambda)$ is an eigenvalue of $r(A)$, with the same eigenvector. No diagonalizability assumption is needed. Equivalently, the polynomial identity

$$r(t) - r(\lambda) = (t - \lambda)s_\lambda(t)$$

evaluated at A gives

$$r(A) - r(\lambda)I = (A - \lambda I)s_\lambda(A),$$

so the eigenvector argument is independent of diagonalization.

Equality of the complete spectral sets also requires the reverse inclusion. Let $A \in \mathbb{C}^{n \times n}$ and let r be nonconstant. If μ is an eigenvalue of $r(A)$, factor over $\mathbb{C}$.

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Polynomial spectral mapping without diagonalization (continued)

$$r(t) - \mu = c \prod_{j=1}^{d} (t - \zeta_j), \qquad c \neq 0.$$

Substitute the matrix and take determinants.

$$0 = \det(r(A) - \mu I) = c^n \prod_{j=1}^{d} \det(A - \zeta_j I).$$

At least one factor vanishes, so some ζ_j is an eigenvalue of A, and $r(\zeta_j) = \mu$. Therefore

$$\operatorname{spec}(r(A)) = r(\operatorname{spec}(A)).$$

For a constant polynomial $r = c$, use $r(A) = cI$ directly. The complete spectrum of a real matrix is also understood over $\mathbb{C}$; listing only real eigenvalues can omit conjugate pairs. This proof gives spectral sets. To count algebraic multiplicities, apply complex Schur triangularization. The diagonal entries of $r(T)$ are $r(T_{ii})$, even when T is not diagonal.

For a nondiagonalizable example, take

Polynomial spectral mapping without diagonalization (continued)

$$A = \begin{bmatrix} 2 & 1 \\ 0 & 2 \end{bmatrix} = 2I + N, \qquad N = \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix},$$

and $r(t) = t^2 + t + 1$. Since $N^2 = 0$,

$$r(A) = r(2)I + r'(2)N = 7I + 5N = \begin{bmatrix} 7 & 5 \\ 0 & 7 \end{bmatrix}.$$

Direct multiplication gives the same matrix, whose eigenvalue is $r(2) = 7$. The derivative term retains the off-diagonal Jordan structure and explains why evaluating a polynomial on eigenvalues alone does not determine the matrix.

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From vectors to equations

Check dimensions first, then use linear combinations to decide which targets are reachable. Dot products describe lengths and orthogonality; elimination finds coefficients systematically in the next chapter.

References and attribution

Lin Cing, handwritten linear algebra notes. Gilbert Strang, Introduction to Linear Algebra, 5th ed., Chapter 1.

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